Output Form (OF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *ASAP++* | ***Context:*** *Qatari and Arab High School Arabic Essay Writers (Grades 10–12)*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark's output form is integer ordinal scores per trait, evaluated by Quadratic Weighted Kappa using an Ordinal Class Classifier with Random Forest internals under 5-fold CV [Q34, Q35, Q36, Q37, Q38, Q39]. The deployment explicitly requires free-text natural-language revision suggestions with explanatory rationale (elicitation Q3/A3). QWK has no mechanism to evaluate the quality, actionability, or pedagogical appropriateness of generated feedback text. This is a complete output-form mismatch: numeric ordinal classification vs. open-ended generation. The benchmark cannot validate the deployment's output modality at all.

ID	Evidence
Q3	“The ASAP AEG dataset comprises of approximately 13,000 essays, written across 8 prompts. The essays were written by students of class 7 to 10. Each essay was evaluated by 2 evaluators. 6 out of the 8 prompts only have overall scores. Only 2 of them have scores for individual essay attributes, like content, organization, style, etc.” (p.1)
Q34	“We evaluate each of the annotators using Cohen's Kappa, with quadratic weights - i.e. Quadratic Weighted Kappa (QWK) (Cohen, 1968).” (p.4)
Q35	“We chose this as the evaluation metric (as compared to accuracy and weighted F-Score) because of the following reasons: Unlike accuracy and F-Score, Kappa takes into account random agreement.” (p.4)
Q36	“Weighted Kappa, takes into account the distance between the actual score and the reported score. Quadratic weights reward matches and penalize mismatches more than linear weights.” (p.4)
Q37	“Due to these reasons, this is one of the most used evaluation metrics to evaluate the performance of essay grading systems. To the best of our knowledge, all of the papers using the ASAP dataset make use of this as the evaluation metric.” (p.4)
Q38	“We made use of the Ordinal Class Classifier (Frank and Hall, 2001) in Weka (Frank et al., 2016). This classifier is a meta-classifier, that first converts ordinal data into categorical data, before running an internal classifier on the data. We used the Random Forest classifier (Breiman, 2001) as the internal classifier.” (p.4)
Q39	“We used 5-fold cross validation to get the results for each attribute for each prompt.” (p.4)

Strengths

- QWK is a methodologically appropriate metric for ordinal-score agreement and is the field standard [Q35, Q36, Q37]; if the deployment ever needs an internal numeric score head (e.g., for triggering feedback templates), the QWK methodology is reusable.

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Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: No — expected output modality is numeric ordinal scores [Q34, Q38], not natural-language feedback. The deployment requires generated Arabic-script revision suggestions with rationale (elicitation Q3/A3).

OF-2: Not applicable directly: text-to-speech is not the deployment's primary output channel, but the deployment does require Arabic-script text generation, which the benchmark does not produce or evaluate at all. RTL Arabic rendering on the Qeducation LMS [WEB-15, WEB-17] would be the relevant infrastructure consideration.

OF-3: Literacy is high in Qatar (general internet penetration ~99–100% suggests strong digital access [WEB-13, WEB-14]); accessibility for Arabic-script text feedback on the Qeducation mobile app is supported [WEB-17]. The benchmark does not address any of this because it produces no text output.

OF-4: The mismatch is total: numeric ordinal score outputs evaluated by QWK [Q34–Q39] cannot validate generated explanatory feedback. External validity for the deployment's output form is zero.

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WEB-13	https://datareportal.com/reports/digital-2026-qatar
WEB-14	https://thepeninsulaqatar.com/article/18/02/2025/at-100-qatar-continues-to-lead-in-global-internet-penetration
WEB-15	Qeducation LMS (CYPHER) supports Arabic-language interface — relevant target platform for natural-language Arabic feedback delivery (https://www.cypherlearning.com/blog/k-20/qatar-march-workshop-recap-empowering-students-teachers-and-parents-with-digital-technology-for-learning)
WEB-17	Official 'Qatar Education' mobile app provides RTL Arabic delivery channel (https://play.google.com/store/apps/details?id=app.qeducation.edu.gov.qa)
WEB-22	Arwi (CAMeLBERT-MSA, ZAEBUC) demonstrates the kind of feedback-generation output form the deployment requires; QWK is insufficient to evaluate such systems (https://arxiv.org/html/2504.11814v1)

Information Gaps

- How feedback-generation quality should be evaluated for Arabic high school writing is itself an open methodological question; no validated framework exists [WEB-22].

Requires Expert Verification

- Pedagogical appropriateness criteria for Arabic feedback text (e.g., register, formality, use of Arabic technical grammar terminology) from Qatari teacher experts.

Remediation

Gap 1: QWK-evaluated numeric scores cannot validate natural-language revision suggestions with explanatory rationale.

Recommendation: Develop a feedback-generation evaluation framework with Qatari teachers (e.g., rubric-aligned ratings of feedback usefulness, accuracy, and pedagogical appropriateness), using Arwi [WEB-22] as a methodological reference, and retain QWK only as a secondary check on any internal numeric score head.

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